

SN Stitches

FASHION · DESIGN · STITCH

System Design & UI Specification
Prepared for Richard Frimpong · October 2026

SN Stitches — System Design

Brand: SN Stitches — *Fashion · Design · Stitch* · Accra, Ghana **Contact:** Instagram @sn_stitches · TikTok @sn_stitches · Phone/WhatsApp 054 692 3596 (`233546923596`) **Market:** mostly Ghana, prices in Ghana cedis (GH₵ / GHS), payments by Mobile Money (MTN, Telecel, AirtelTigo) and card. **Visual design (source of truth for UI):** high-fidelity mockups in `ui-design/` (`screens/desktop`, `screens/mobile`, `SN-Stitches-UI-Design.pdf`) + `UI_SPEC.md`. `reference-prototype/` is an older functional prototype kept only for flows and copy. **Deployment:** Richard deploys to Vercel himself (§14a). The build agent must not deploy.

Legend: **[P1]** = Phase 1 / MVP · **[P2]** = later · **[P3]** = nice-to-have

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1. Goals & constraints

- Sell ~10–200 dresses (sizes XS–XXL, a few colours each) to customers on mobile phones, often on slow or expensive mobile data.
- Payments must be **verified on the server** (Paystack), never trusted from the browser.
- One owner/admin (maybe 1–2 staff) manages products, stock and orders from a phone.
- Running costs close to **GH₵0/month** at launch (free tiers). Simple enough for one freelance developer to maintain.

2. Recommended stack (and why)

Layer	Choice	Why
Framework	**Next.js 15 (App Router) + TypeScript**	Server Components give small JS bundles (good on mobile data). Server Actions and Route Handlers mean there's no separate backend. Strong SEO support. First-class on Vercel.
Styling	**Tailwind CSS v4** + headless primitives (Radix / shadcn/ui)	Tokens live in one config. Primitives are accessible. The "Atelier" signature styles (UI_SPEC §0) are custom CSS utilities.
Database	**Prisma Postgres** (managed Postgres by Prisma)	Free plan, no credit card. Built-in managed connection pool (PgBouncer) that suits serverless functions. One vendor for the DB and the ORM.
ORM / migrations	**Prisma ORM** (current major)	Type-safe queries, `prisma migrate` for versioned migrations, `prisma db seed`, Prisma Studio for quick data fixes.
Admin auth	**Auth.js (NextAuth v5)** + `@auth/prisma-adapter`	Email magic link (via Resend) and/or credentials (argon2/bcrypt hash). Sessions stored in Postgres. A `role` field on `User` gates the admin.
Images	**Vercel Blob** (public store)	Admin uploads straight from the product editor (client upload with a server-issued token). Served from the CDN. Optimised with `next/image`.
Payments	**Paystack** (Popup/Inline + Transactions API + webhooks)	Ghana-native. GHS, MoMo (MTN, Telecel, AirtelTigo) and cards.
Email	**Resend** + React Email	Owner notifications, customer receipts and Auth.js magic links.
Hosting	**Vercel**	Zero-config Next.js, preview deploys, Blob and Analytics built in. **Richard deploys it himself. The build agent never deploys.**

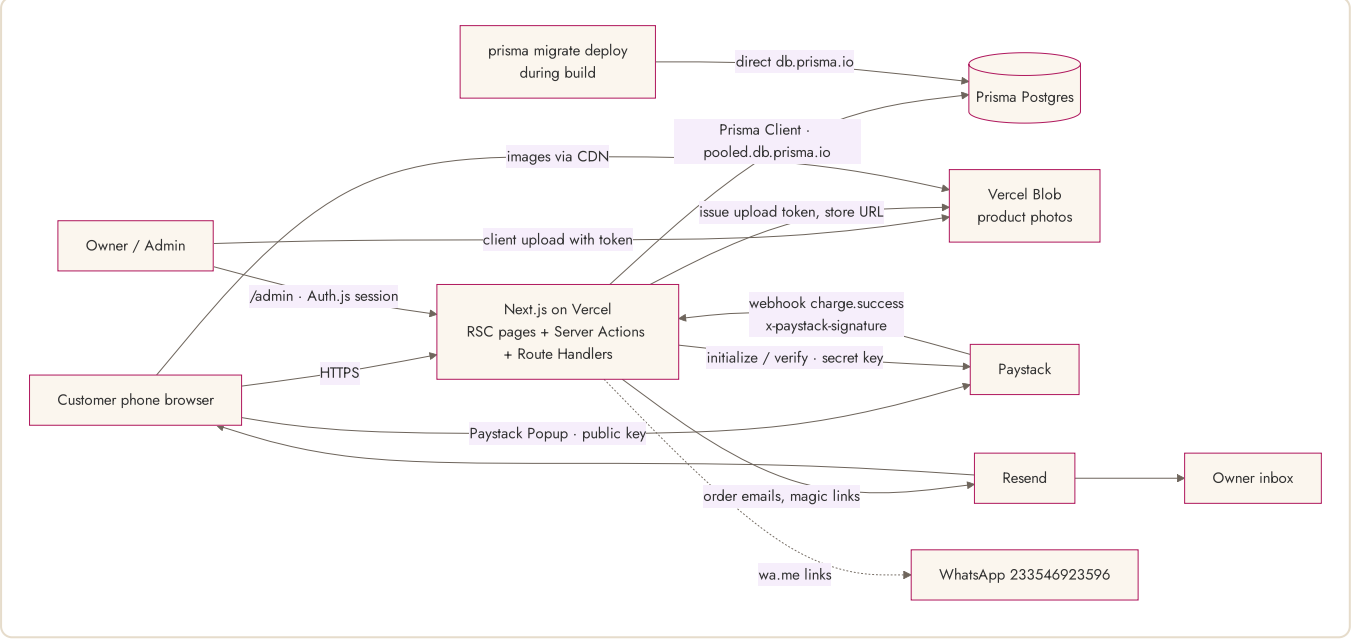
Layer	Choice	Why
Validation	**Zod**	One schema shared by forms and server actions.
Cart state	Zustand + `persist` (localStorage)	Instant, works offline, no DB operations spent on browsing.
Testing	Vitest (unit), Playwright (e2e)	Protects checkout and payment logic.

Free-tier facts (checked on the vendors' pages on 29 Oct 2026; re-check before launch): - **Prisma Postgres Free:** $0, no card. **200,000 operations / month** (each ORM query = 1 operation) shared across the workspace. Up to 50 databases. **10 pooled + 10 direct connections** (5 direct are reserved for the platform). **No automated backups** on Free. When the operation limit is reached, queries are **blocked until the next calendar month** (one-time 24h grace) unless you upgrade. Free storage is listed as 500 MB on the pricing page and 2 GB in the docs, so confirm in the console. Starter is $10/month (1M operations, 10 GB, daily backups). - **Vercel Blob (Hobby):** 1 GB storage, 10K simple operations (cache-miss reads), 2K advanced operations (uploads/lists) and 10 GB data transfer per month. Access is paused (not billed) if a limit is exceeded. - **Vercel Hobby is for non-commercial, personal use** under Vercel's fair-use terms. A live shop taking payments should move to **Vercel Pro (about $20/month)** at launch. Hobby is fine for previews.

Keeping within 200k operations/month: catalogue pages are statically generated / ISR with `revalidateTag('products')` on admin save, so browsing makes almost no DB queries. The cart lives on the device. Checkout uses about 10–15 operations per order. Avoid polling loops (the confirmation poll is capped at 20 calls). Add `npx prisma postgres usage` / console checks to the go-live checklist. Upgrade to Starter if traffic grows.

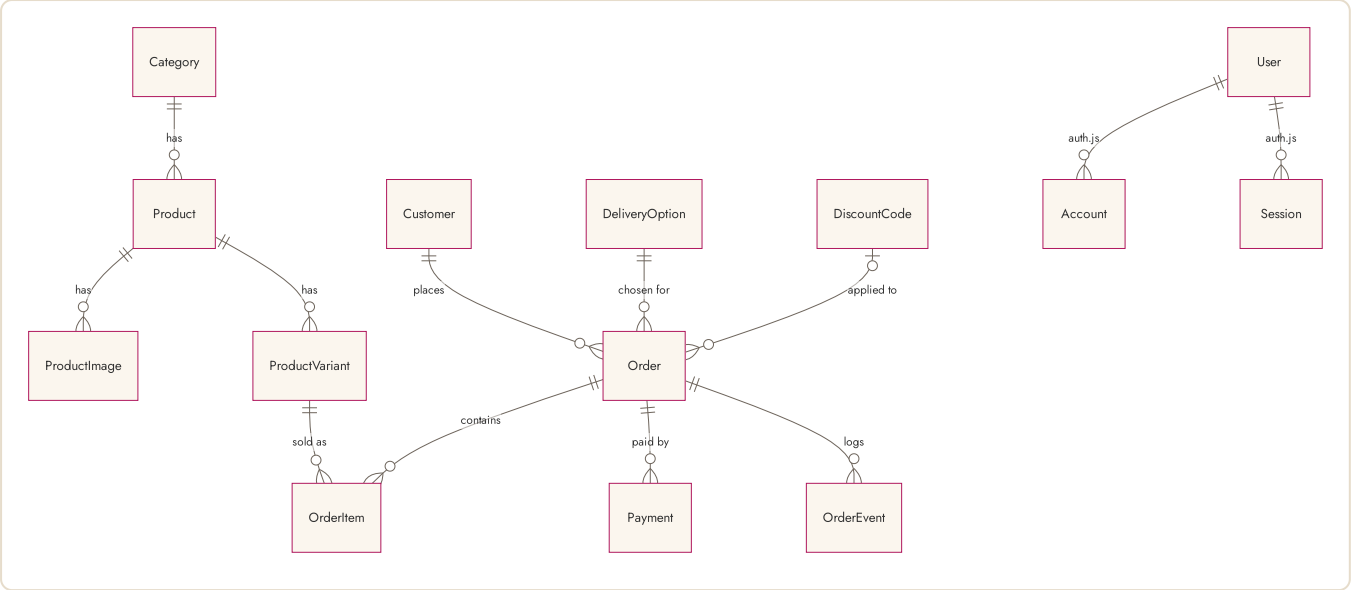
Rejected for now: Shopify (monthly USD fees, weaker MoMo/GHS fit), Supabase (replaced at Richard's request by Prisma Postgres), a custom Node API.

3. Architecture



4. Data model (Prisma Postgres)

Money is **Int pesewas** everywhere (GH₵450.00 = `45000`). IDs are `cuid()`. Timestamps use `@default(now())` / `@updatedAt`.



`prisma/schema.prisma` (draft)

```
// SN Stitches: Prisma schema draft. Money = Int pesewas.
// Runtime traffic uses the POOLED url (pooled.db.prisma.io); CLI/migrations use the DIRECT
url (db.prisma.io).
// Follow the current Prisma ORM docs for where the URLs live (schema datasource vs
prisma.config.ts)
// and whether a driver adapter (e.g. @prisma/adapter-pg) is required in your Prisma major
version.

generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider  = "postgresql"
  url       = env("DATABASE_URL")   // pooled
  directUrl = env("DIRECT_URL")     // direct, for migrate/studio
}

enum Role            { CUSTOMER STAFF OWNER }
enum ProductStatus   { DRAFT ACTIVE ARCHIVED }
enum DressLength     { MINI MIDI MAXI PLAYSUIT }
enum Size            { XS S M L XL XXL }
enum OrderStatus     { PENDING PAID PROCESSING SHIPPED DELIVERED CANCELLED REFUNDED }
enum PaymentMethod   { MOMO CARD WHATSAPP }
enum PaymentChannel  { MOBILE_MONEY CARD OTHER }
enum PaymentStatus   { INITIALIZED SUCCESS FAILED ABANDONED }
enum MomoNetwork     { MTN TELECEL AIRTELTIGO }
enum DiscountType    { PERCENT FIXED }
enum GhanaRegion {
  GREATER_ACCRA ASHANTI CENTRAL EASTERN WESTERN WESTERN_NORTH VOLTA OTI
  NORTHERN SAVANNAH NORTH_EAST UPPER_EAST UPPER_WEST BONO BONO_EAST AHAFO
}
```

```
model Category {
  id          String    @id @default(cuid())
  name        String
  slug        String    @unique
  description String?
  imageUrl    String?
  sortOrder   Int       @default(0)
  isActive    Boolean   @default(true)
  products    Product[]
  createdAt   DateTime  @default(now())
  updatedAt   DateTime  @updatedAt
}

model Product {
  id               String           @id @default(cuid())
  name             String
  slug             String           @unique
  description      String
  details          String[]
  length           DressLength
  pricePesewas     Int
  compareAtPesewas Int?
  status           ProductStatus    @default(DRAFT)
  isFeatured       Boolean          @default(false)
  madeToMeasure    Boolean          @default(false)
  mtmFeePesewas    Int?             // e.g. 8000 = GH₵80
  leadTimeDays     Int?
  seoTitle         String?
  seoDescription   String?
  isSample         Boolean          @default(false) // seed data marker
  categoryId       String
  category         Category         @relation(fields: [categoryId], references: [id])
  images           ProductImage[]
  variants         ProductVariant[]
  orderItems       OrderItem[]
  createdAt        DateTime         @default(now())
  updatedAt        DateTime         @updatedAt
  @@index([status, isFeatured])
  @@index([categoryId])
}

model ProductImage {
  id        String  @id @default(cuid())
  productId String
  product   Product @relation(fields: [productId], references: [id], onDelete: Cascade)
  url       String  // Vercel Blob public URL
  pathname  String  // Blob pathname (for delete)
  alt       String
  width     Int?
  height    Int?
  sortOrder Int     @default(0) // 0 = main, 1 = hover image
  @@index([productId, sortOrder])
}

model ProductVariant {
  id                    String      @id @default(cuid())
  productId             String
```

```
  product              Product     @relation(fields: [productId], references: [id],
onDelete: Cascade)
  size                 Size
  colourName           String
  colourHex            String?
  sku                  String      @unique
  stockQty             Int         @default(0) // CHECK (stock_qty >= 0) added in SQL
migration
  priceOverridePesewas Int?
  isActive             Boolean     @default(true)
  orderItems           OrderItem[]
  @@unique([productId, size, colourName])
}

model Customer {
  id             String   @id @default(cuid())
  fullName       String
  phone          String   @unique // E.164 +233…
  email          String?
  marketingOptIn Boolean  @default(false)
  orders         Order[]
  createdAt      DateTime @default(now())
}

model DeliveryOption {
  id          String   @id @default(cuid())
  code        String   @unique // ACCRA_SAME_DAY, ACCRA_NEXT_DAY, REGIONAL, STUDIO_PICKUP
  label       String
  etaText     String
  feePesewas  Int
  regions     GhanaRegion[] // empty = all regions
  cutoffTime  String?  // "12:00" Africa/Accra
  isPickup    Boolean  @default(false)
  isActive    Boolean  @default(true)
  sortOrder   Int      @default(0)
  orders      Order[]
}

model Order {
  id                 String        @id @default(cuid())
  orderNumber        String        @unique // SNS-24001…
  accessToken        String        @unique @default(cuid()) // for /order/[number]?t=
  status             OrderStatus   @default(PENDING)
  needsAttention     Boolean       @default(false)
  paymentMethod      PaymentMethod
  momoNetwork        MomoNetwork?
  customerId         String
  customer           Customer      @relation(fields: [customerId], references: [id])
  contactName        String
  contactPhone       String
  contactEmail       String
  region             GhanaRegion
  city               String
  gpsAddress         String?
  addressLine        String
  landmark           String?
  deliveryOptionId   String
```

```
  deliveryOption     DeliveryOption @relation(fields: [deliveryOptionId], references: [id])
  deliveryFeePesewas Int
  subtotalPesewas    Int
  discountPesewas    Int           @default(0)
  totalPesewas       Int
  discountCodeId     String?
  discountCode       DiscountCode? @relation(fields: [discountCodeId], references: [id])
  notes              String?
  paidAt             DateTime?
  stockCommitted     Boolean       @default(false)
  notifiedAt         DateTime?     // notifications sent once
  utm                Json?
  items              OrderItem[]
  payments           Payment[]
  events             OrderEvent[]
  createdAt          DateTime      @default(now())
  updatedAt          DateTime      @updatedAt
  @@index([status, createdAt])
}

model OrderItem {
  id                String         @id @default(cuid())
  orderId           String
  order             Order          @relation(fields: [orderId], references: [id], onDelete:
Cascade)
  productId         String
  product           Product        @relation(fields: [productId], references: [id])
  variantId         String
  variant           ProductVariant @relation(fields: [variantId], references: [id])
  productName       String  // snapshot
  size              Size
  colourName        String
  unitPricePesewas  Int
  qty               Int
  madeToMeasure     Boolean @default(false)
  lineTotalPesewas  Int
}

model Payment {
  id            String         @id @default(cuid())
  orderId       String
  order         Order          @relation(fields: [orderId], references: [id])
  provider      String         @default("paystack")
  reference     String         @unique
  amountPesewas Int
  currency      String         @default("GHS")
  channel       PaymentChannel?
  status        PaymentStatus  @default(INITIALIZED)
  raw           Json?
  verifiedAt    DateTime?
  createdAt     DateTime       @default(now())
}

model OrderEvent {
  id         String       @id @default(cuid())
  orderId    String
  order      Order        @relation(fields: [orderId], references: [id], onDelete: Cascade)
```

```
  fromStatus OrderStatus?
  toStatus   OrderStatus?
  note       String?
  actor      String       // "system" | "webhook" | admin email
  createdAt  DateTime     @default(now())
}

model DiscountCode { // Phase 2
  id                 String       @id @default(cuid())
  code               String       @unique
  type               DiscountType
  value              Int          // percent (0–100) or pesewas
  minSubtotalPesewas Int?
  startsAt           DateTime?
  endsAt             DateTime?
  maxUses            Int?
  usedCount          Int          @default(0)
  isActive           Boolean      @default(true)
  orders             Order[]
}

model StoreSettings { // single row, id = "default"
  id                          String  @id @default("default")
  freeDeliveryThresholdPesewas Int     @default(150000)
  whatsappNumber              String  @default("233546923596")
  contactPhone                String  @default("0546923596")
  instagram                   String  @default("sn_stitches_")
  tiktok                      String  @default("sn_stitches")
  contactEmail                String?
  announcementText            String?
}

// ---------- Auth.js (NextAuth v5) Prisma adapter models ----------
model User {
  id            String    @id @default(cuid())
  name          String?
  email         String    @unique
  emailVerified DateTime?
  image         String?
  passwordHash  String?   // only if the credentials provider is enabled
  role          Role      @default(CUSTOMER)
  accounts      Account[]
  sessions      Session[]
  createdAt     DateTime  @default(now())
}
model Account {
  userId            String
  type              String
  provider          String
  providerAccountId String
  refresh_token     String?
  access_token      String?
  expires_at        Int?
  token_type        String?
  scope             String?
  id_token          String?
  session_state     String?
```

```
  user              User    @relation(fields: [userId], references: [id], onDelete: Cascade)
  @@id([provider, providerAccountId])
}
model Session {
  sessionToken String   @unique
  userId       String
  expires      DateTime
  user         User     @relation(fields: [userId], references: [id], onDelete: Cascade)
}
model VerificationToken {
  identifier String
  token      String
  expires    DateTime
  @@id([identifier, token])
}
```

Order status: `PENDING` → `PAID` → `PROCESSING` → `SHIPPED` (out for delivery / dispatched) → `DELIVERED`. `CANCELLED` is allowed from pending, paid or processing (and restocks). `REFUNDED` is P2. Allowed transitions are enforced in one `canTransition(from,to)` helper. **Stock** is decremented when the order becomes `PAID`, inside a `prisma.$transaction` using a conditional `updateMany({ where:{ id, stockQty:{ gte: qty } }, data:{ stockQty:{ decrement: qty } } })`. If the count is 0, the order is flagged `needsAttention`.

Migrations & seed

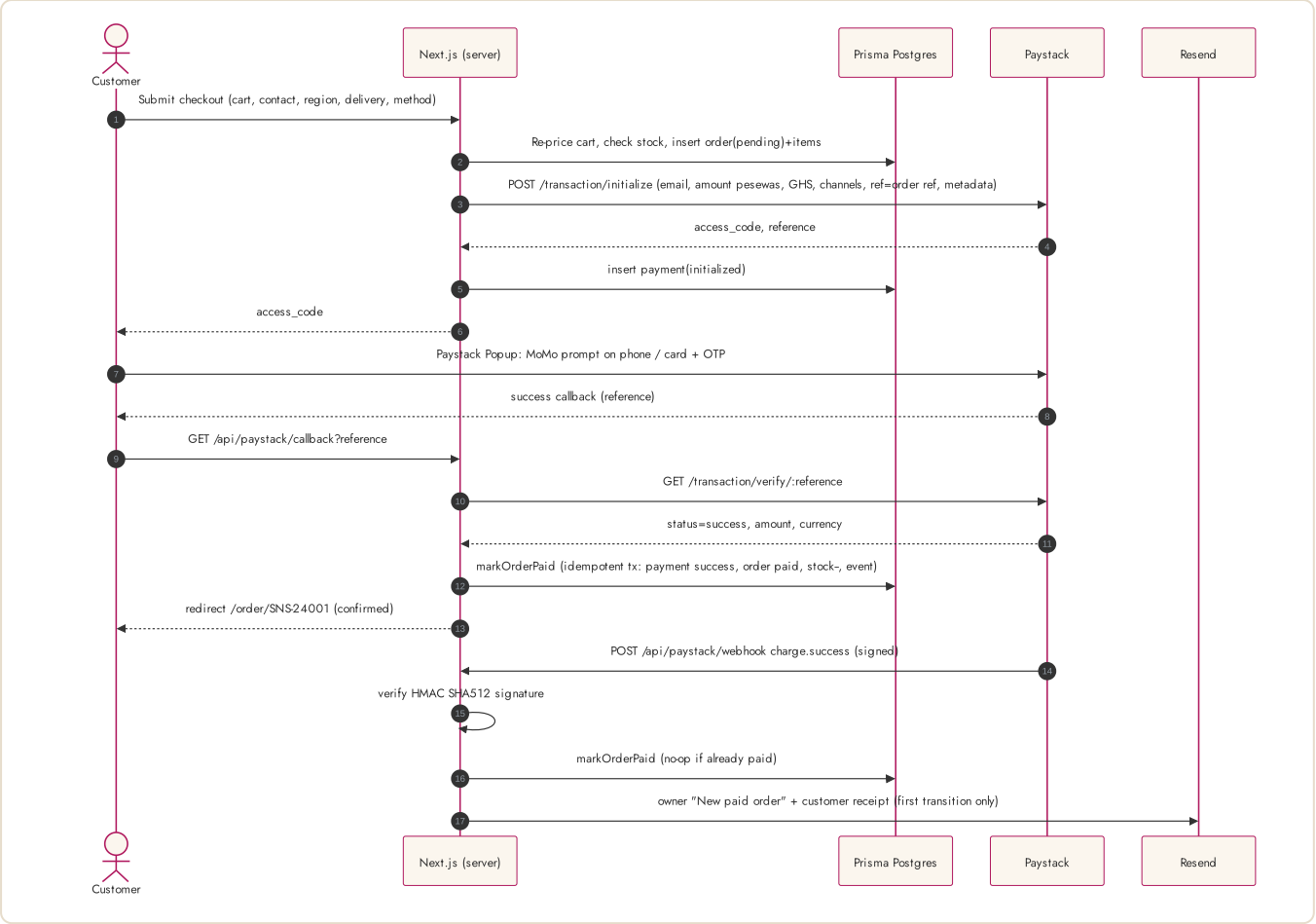
- `prisma migrate dev --name init` locally. Commit `prisma/migrations/**`. Add a raw-SQL follow-up migration for `CHECK (stock_qty >= 0)` and a sequence for `orderNumber` (`SNS-24001` upward, or a counter row updated in a transaction).
- **Production:** Vercel build command `prisma migrate deploy && next build` (migrations use `DIRECT_URL`). `postinstall: prisma generate`.
- **Seed** (`prisma/seed.ts`, run with `prisma db seed`, idempotent upserts):
- Categories (Ankara, Kente, Evening, Casual).
- ~12 **sample** products from `reference-prototype/products.js` plus the **Sika Tie-Dye Playsuit**, all `isSample: true`, with variants XS–XXL (stock 0–8, at least one sold-out size).
- Images: in dev, serve from `/public/sample/*.webp`. With `BLOB_READ_WRITE_TOKEN` set, upload them to Vercel Blob via `put()` and store the URLs.
- The 4 delivery options and `StoreSettings`.
- An OWNER user from `SEED_ADMIN_EMAIL` (plus a `SEED_ADMIN_PASSWORD` hash if credentials are enabled).

5. Server actions & API routes

Kind	Path / name	Phase	Purpose
RSC page	`/`, `/shop`, `/shop/[category]`, `/product/[slug]`, `/about`, `/contact`, `/size-guide`, `/faq`, `/checkout`, `/order/[orderNumber]`	P1	storefront (cart is client state)
Server action	`validateCart(items)`	P1	re-prices the cart from the DB and returns availability (never trust client prices)

Kind	Path / name	Phase	Purpose
Server action	`getDeliveryOptions(region)`	P1	options and fees for a region (+ free-over-threshold)
Server action	`createOrder(checkoutInput)`	P1	Zod validate → re-price → create customer, order `pending` and items → Paystack `transaction/initialize` (amount in pesewas, GHS, channels, metadata.order_id, callback_url) → return `{access_code, reference}` (method=whatsapp: return wa.me URL, status stays `pending`)
Route handler	`GET /api/paystack/callback?reference=`	P1	after the popup/redirect: `transaction/verify/:reference` → `markOrderPaid()` (idempotent) → redirect to `/order/[orderNumber]`
Route handler	`POST /api/paystack/webhook`	P1	HMAC-SHA512 signature check (`x-paystack-signature`, secret key) on the raw body → on `charge.success` re-verify the amount and currency against the order → `markOrderPaid()` (idempotent) → 200 quickly
Lib (server)	`markOrderPaid(reference, payload)`	P1	one transaction: payment success, order `paid`, decrement stock, event log; then send notifications (only on the first transition)
Route handler	`GET /api/orders/[orderNumber]/status`	P1	confirmation page polls while status=pending (webhook may land after redirect). Returns status only, keyed by order number + short token
Route handler	`POST /api/admin/blob-upload`	P1	Vercel Blob **client-upload token** route (`handleUpload` from `@vercel/blob/client`). `onBeforeGenerateToken` calls `requireAdmin()` and restricts content types (jpeg/png/webp) and max size (8 MB). `onUploadCompleted` is optional (on localhost, save the URL from the client instead)
Server actions (admin)	`upsertProduct`, `archiveProduct`, `addProductImage(url, pathname, alt)`, `deleteProductImage` (Blob `del()`), `reorderImages`, `upsertVariant/adjustStock`, `updateOrderStatus(orderId, to, note)`, `updateSettings`, `upsertDeliveryOption`, `upsertDiscount` [P2]	P1	**every action starts with `await requireAdmin()`** (Auth.js `auth()` + role OWNER/STAFF)
Route handler	`/api/auth/[...nextauth]`	P1	Auth.js v5 handlers
Route handler	`GET /api/admin/orders.csv`	P2	export
Route	`/sitemap.xml`, `/robots.txt`, OG image route	P1	SEO

6. Payment flow



Rules: the amount and currency must equal `orders.total_pesewas` /GHS or the order is flagged. The reference is unique per attempt. If the popup closes, the order stays `pending` and the customer can retry. Pending orders older than 24h are auto-cancelled by a scheduled job [P2] (Vercel Cron), or the owner cancels them manually [P1]. **Demo/dev:** use Paystack **test keys** (`pk_test_` / `sk_test_`). If keys are missing, the UI shows a "Payments not configured" banner and blocks payment. No fake success in production builds. An explicit `NEXT_PUBLIC_DEMO_PAYMENTS=true` may simulate success locally only.

7. Notifications

- **[P1] Owner email** (Resend) on a paid order: order number, items/sizes, total, delivery option, customer name, phone, address, landmark, plus a **"Chat customer on WhatsApp"** `wa.me/<customer>` link and an "Open in admin" link.
- **[P1] Customer receipt email**, with order number, items, total, delivery ETA and the SN Stitches WhatsApp link (`wa.me/233546923596`).
- **[P1] WhatsApp click-to-chat** (no API): "Order via WhatsApp" in the cart/checkout builds a prefilled `https://wa.me/233546923596?text=…`. These orders are saved as `pending`, method `whatsapp`, so the owner sees them in admin.
- **[P2]** Status-change emails (shipped/delivered). SMS via Hubtel/Arkesel. WhatsApp Cloud API template messages.

8. Admin dashboard (`/admin`)

- **[P1] Auth: Auth.js (NextAuth v5)** with the Prisma adapter. Email magic link via Resend (recommended), plus an optional credentials provider (email + argon2/bcrypt password). Only users with `role` OWNER or STAFF can enter. `middleware.ts` redirects unauthenticated or non-admin users away from `/admin/**`, and every admin server action and route re-checks with `requireAdmin()`.
- **[P1] Image upload:** the product editor uploads photos **directly from the browser to Vercel Blob** (`upload()` from `@vercel/blob/client` with a token from `/api/admin/blob-upload`). Images are resized client-side to ≤1600px WebP first. The returned URL/pathname is saved as a `ProductImage`. Deleting an image removes the blob with `del()`.
- **[P1] Dashboard:** today's and the week's revenue, orders by status, low-stock variants (≤2), latest 10 orders.
- **[P1] Products:** list (search, filter by category/status), create/edit/archive. Image upload (drag/drop, multiple, reorder, alt text; resized client-side to ≤1600px WebP before upload). Variants grid: rows = sizes XS–XXL, columns = colours, cell = stock qty plus an active toggle. Price, compare-at, featured, made-to-order, SEO fields. "Sample" badge on seed items.
- **[P1] Orders:** table with status tabs (Pending, Paid, Processing, Shipped, Delivered, Cancelled), search by order number/phone/name, date filter. **Order detail:** items, customer, address/landmark/GPS, payment info (Paystack reference, channel), status pipeline buttons with allowed transitions only, notes, timeline (order_events), "WhatsApp customer" and "Call" buttons, printable packing slip.
- **[P1] Settings:** free-delivery threshold, delivery options/fees, WhatsApp/phone/socials, announcement bar.
- **[P2]** Discount codes, CSV export, staff roles, sales charts, customer list.

9. Delivery rules

- All **16 regions:** Greater Accra, Ashanti, Central, Eastern, Western, Western North, Volta, Oti, Northern, Savannah, North East, Upper East, Upper West, Bono, Bono East, Ahafo.
- **Greater Accra:** Same-day (order before 12:00 Africa/Accra, Mon–Sat) GH₵40 · Next-day GH₵25.
- **Other regions:** courier / VIP-bus parcel, 2–4 working days, GH₵55 (configurable; P2: per-zone fees, e.g. Northern zone higher).
- **Studio pickup (Accra):** free, any region.
- **Free delivery** when subtotal (after discount) ≥ **GH₵1,500** (setting). Doesn't apply to pickup.
- Same-day is hidden after the cutoff. Made-to-order items show "ships in N days" and disable same-day.
- Address fields: region (required), city/town (required), GhanaPost GPS (optional, format `XX-000-0000`), address + landmark (required), phone (required, Ghana format → stored E.164).
- Fees are all prototype values, so they're **assumptions to confirm** with the owner.

10. Security

- **No database access from the browser.** Prisma runs only on the server (Server Components, Server Actions, Route Handlers). `DATABASE_URL` is never exposed. There is no RLS, so **authorization is enforced in code:**
- `requireAdmin()` (Auth.js `auth()` + role OWNER/STAFF) is the **first line of every admin server action and admin route handler**, as well as in `middleware.ts` for `/admin/**`. Add a unit test that each admin action rejects anonymous and CUSTOMER sessions.
- Public data access goes through a small `lib/catalog.ts` that only selects `ACTIVE` products, active variants and safe fields (no cost or internal notes).
- Order lookups require `orderNumber` + `accessToken`.

- Server actions validate input with Zod and are rate-limited where abuse is likely (createOrder, magic-link requests, credentials login).
- **Webhook:** verify `x-paystack-signature` = HMAC-SHA512(raw body, `PAYSTACK_SECRET_KEY`), use a constant-time compare, and reject otherwise. Optionally allow-list Paystack IPs. Idempotent by reference.
- **Never trust the client:** prices, fees, totals and stock are recomputed on the server. Zod on every input.
- **Admin:** Auth.js sessions in the DB (revocable). `AUTH_SECRET` set. Magic links expire in 10 minutes. Credentials use argon2/bcrypt with a login rate limit. TOTP 2FA [P2].
- Secrets only in env vars. The only public ones are `NEXT_PUBLIC_PAYSTACK_PUBLIC_KEY`, site URL, WhatsApp/socials and analytics flags.
- Storage: the Vercel Blob store is public read (product photos only, never customer data). Upload tokens are only issued to admins, with content-type and size limits. Security headers (CSP allowing js.paystack.co / checkout.paystack.com), HTTPS only.
- Order page access by `orderNumber` + an unguessable `access_token` in the URL. Don't expose PII to enumeration.
- Ghana Data Protection Act 2012: privacy policy page, minimal PII, marketing opt-in.

11. SEO

- Metadata API per page. Product JSON-LD (`Product`, `Offer` with `priceCurrency: GHS`, availability), `Organization` with `sameAs` IG/TikTok, `BreadcrumbList`.
- `sitemap.xml` (products, categories, static pages), `robots.txt`, canonical URLs, Open Graph/Twitter images (product main image; generated OG for home).
- Clean slugs: `/product/adwoa-ankara-wrap-midi`, `/shop/ankara`. `lang="en-GH"`. Descriptive alt text.
- [P2] Google Merchant Center feed, Instagram/Facebook catalogue feed.

12. Analytics

- [P1] Vercel Analytics or Plausible: page views and custom events `view_item`, `add_to_cart`, `begin_checkout`, `purchase` (fired server-confirmed on the confirmation page), `whatsapp_order_click`.
- [P2] Meta Pixel + TikTok Pixel (consent banner), UTM capture into `orders.utm` jsonb.

13. Performance (mobile data)

- Budget: **< 150 KB JS** on product/shop pages, LCP < 2.5s on mid-range Android / 4G, CLS < 0.05.
- RSC by default. Client components only for the cart, gallery, filters and checkout. Paystack script loaded **only on checkout**, on demand.
- `next/image` with `sizes`, AVIF/WebP, blur placeholders, lazy below the fold. Hero ≤ 120 KB.
- `next/font` for Cormorant Garamond, Jost, Great Vibes (subset latin, `display: swap`, only the weights used).
- ISR / `revalidateTag('products')` after admin edits. Static generation for product pages.
- Cart in `localStorage` (Zustand persist), so there's no network round-trip.

14. Environment variables

Var	Scope	Notes
`NEXT_PUBLIC_SITE_URL`	public	e.g. http://localhost:3000
`DATABASE_URL`	**server only**	Prisma Postgres **pooled** connection string (`pooled.db.prisma.io`), used at runtime
`DIRECT_URL`	**server only**	Prisma Postgres **direct** connection string (`db.prisma.io`), used for `prisma migrate deploy`, Studio, `pg_dump`
`AUTH_SECRET`	**server only**	`npx auth secret`
`AUTH_URL`	server	optional on Vercel (auto-detected). Set to the site URL elsewhere
`AUTH_RESEND_KEY`	server only	Auth.js Resend provider (can equal `RESEND_API_KEY`)
`ADMIN_EMAILS`	server only	comma list allowed to become OWNER on first sign-in (bootstrap)
`BLOB_READ_WRITE_TOKEN`	**server only**	Vercel Blob store token (added automatically when a Blob store is connected to the Vercel project)
`SEED_ADMIN_EMAIL` / `SEED_ADMIN_PASSWORD`	local only	used by `prisma db seed`
`NEXT_PUBLIC_PAYSTACK_PUBLIC_KEY`	public	pk_test_... / pk_live_...
`PAYSTACK_SECRET_KEY`	**server only**	sk_...; also used for webhook HMAC
`RESEND_API_KEY`	server only	
`ORDER_NOTIFY_EMAIL`	server only	owner's inbox
`EMAIL_FROM`	server only	e.g. `SN Stitches <orders@yourdomain>`
`NEXT_PUBLIC_WHATSAPP_NUMBER`	public	`233546923596`
`NEXT_PUBLIC_INSTAGRAM` / `NEXT_PUBLIC_TIKTOK`	public	`sn_stitches_` / `sn_stitches` (fallbacks; DB settings win)
`NEXT_PUBLIC_DEMO_PAYMENTS`	public	`true` only for local demo; ignored when `NODE_ENV=production`
`CRON_SECRET`	server only	[P2] pending-order cleanup

Ship a `.env.example` with all of these and no real values.

14a. Vercel deployment checklist (Richard runs this, the build agent does not deploy)

1. **Prisma Postgres:** create a database in the Prisma Console (or via the Vercel Marketplace integration). Copy the **pooled** URL to `DATABASE_URL` and the **direct** URL to `DIRECT_URL`.
2. **Vercel project:** import the Git repo. Framework: Next.js. **Build command:** `prisma migrate deploy && next build` (plus `postinstall: prisma generate` in package.json). Node 20+.

3. **Vercel Blob:** Storage → Create Blob store (public) → connect to the project. This adds `BLOB_READ_WRITE_TOKEN`.
4. **Env vars** (Production + Preview): everything in §14. Use Paystack **test** keys on Preview and **live** keys on Production only after testing.
5. **Paystack dashboard** → Settings → API Keys & Webhooks:
6. Webhook URL `https://<domain>/api/paystack/webhook`
7. Callback URL `https://<domain>/api/paystack/callback`
8. Enable GHS, Mobile Money and Card.
9. **Resend:** verify the sending domain (SPF/DKIM DNS). Set `EMAIL_FROM` and `ORDER_NOTIFY_EMAIL`.
10. **First deploy**, then run the seed **once** against production only if you want sample data (`DATABASE_URL=… npx prisma db seed`). Otherwise create real products in /admin and delete the samples.
11. **Sign in** at `/admin` with an email listed in `ADMIN_EMAILS`. Confirm OWNER access.
12. **Test:**
13. Pay a GH₵1 test product with MoMo and with a card in test mode.
14. Confirm the order becomes PAID via the webhook, stock goes down once, and both emails arrive.
15. Replay the webhook and check nothing changes.
16. **Go live:** switch to live Paystack keys, add the custom domain, enable Vercel Analytics, upgrade to Vercel Pro (commercial use), and check the Prisma usage page weekly in the first month.

15. Phasing

Phase 1 — MVP: storefront (all pages), cart, checkout with Paystack MoMo + card and server verification + webhook, WhatsApp ordering, owner + customer emails, admin (auth, products + images + stock per size, orders pipeline, settings, delivery options), SEO basics, analytics basics, seed sample data. **Phase 2:** discount codes, customer accounts and order history, status emails/SMS, pending-order auto-cancel cron, CSV export, Meta/TikTok pixels and catalogue feeds, wishlists, reviews, per-zone regional fees, made-to-measure measurement form. **Phase 3:** WhatsApp Cloud API, multi-currency for diaspora (USD/GBP via Paystack), loyalty, lookbook/blog CMS.

16. Cost estimate (launch)

Vercel Hobby $0 for previews (non-commercial use only), so use **Vercel Pro ≈ $20/mo once the shop is live**. Prisma Postgres Free $0 (200k operations/month; Starter $10/mo adds backups and 1M operations). Vercel Blob within the plan allowance. Resend Free (3,000 emails/month, 100/day) · Paystack: no monthly fee, a per-transaction fee (check current Ghana rates on paystack.com) · domain ≈ $10–15/yr.

Admin: create listing (draft vs publish)

- **Route:** `/admin/products/new` (server component plus client stepper). Uploads go through `/api/admin/blob-upload`.
- **`saveDraft(input)`** (server action, `requireAdmin()`): creates or updates a Product with `status: DRAFT`. Only the name is required; called by autosave.
- **`createProduct(input, { publish: boolean })`** (server action, `requireAdmin()`): validates with Zod.
- **Draft:** name only.
- **Publish:** ≥1 ProductImage with alt text, price > 0, compare-at > price if set, category, ≥1 ProductVariant with stock > 0.

- Checks slug uniqueness (appends `-2` etc.).
- Writes Product + images + variants in one `prisma.$transaction`, sets `ACTIVE` when publishing, then `revalidatePath('/shop')` and the product path.
- Returns `{ id, slug, url }` for the success screen.
- **Share actions** run client-side only (WhatsApp `wa.me/?text=`, copy link and caption, download the main image for IG/TikTok). No social APIs.

SN Stitches — UI Specification

▸ The mockups are the source of truth

High-fidelity screens live in `ui-design/`. **Match them pixel-closely** (layout, spacing, type sizes, colours, motifs, copy) at 1440px and 390px. Interpolate sensibly in between. - `ui-design/screens/desktop/NN-name-desktop.png` and `ui-design/screens/mobile/NN-name-mobile.png`. Full-page renders: floating bars (mobile sticky add-to-bag, admin save bar, WhatsApp FAB) are drawn at the end of the page. In the app they are fixed to the viewport. - `ui-design/html/*.html` + `ui-design/ds.css` are the exact HTML/CSS sources of every mockup. **Open them in a browser and inspect them for exact values.** Port `ds.css` into Tailwind tokens and utilities, not the other way round. - `ui-design/SN-Stitches-UI-Design.pdf` is the presentation. `ui-design/boards/*.png` are the overview boards.

#	Screen	#	Screen
00	Design system	07	Order confirmation
01	Home	08	Our story / About
02	Shop with filters	09	Contact
03	Product detail (Sika Tie-Dye Playsuit)	10	Size guide & FAQ
04	Cart drawer	11	Empty bag
05	Checkout: delivery	12	404
06	Checkout: payment (MoMo network choice)	21–28	Admin: login, dashboard, products, product editor, orders, order detail

Older functional prototype (flows only, superseded visually): `reference-prototype/`.

0. Signature visual language: "Atelier"

The store must **not** look like a generic Shopify-style template. Its identity comes from the logo and the craft of tailoring. Every screen should carry at least two of these motifs, used with restraint:

Motif	Spec	Where
Stitch seams	`1.5px dashed #B0185E` (55–80% opacity). Cards use an **inset stitched outline** (`::after`, inset 7px, inherits radius)	section dividers, accordions, drawer header, checkout sticky bars, featured cards, focus state of inputs (`1.5px dashed` magenta + 3px magenta-50 ring)
Thread line	SVG path, magenta, `stroke-dasharray: 7 6`, 1.6px, weaving through the hero and between sections. **Motion:** draws on scroll (`stroke-dashoffset` animation, 1.2s ease-out, once). Disabled with `prefers-reduced-motion`	home hero, process section, about hero, 404
Gold foil	1px `#D4A437` border + inner 3px cream gap + 1px gold hairline (`.foil`). Primary buttons have an inner gold hairline. Gold-gradient button on dark backgrounds	hero/gallery photos, primary CTA, dark sections

Motif	Spec	Where
Linen texture	cream page with two ultra-faint repeating line gradients (≈3–5% opacity). `.linen` = sand version. CSS only, no image download	page background, footer, quotes, how-to-measure
Kente band	repeating gold/ink/magenta/green stripe, 4px (thin) or 8px	under announcement bar, above footer, receipt hem, section breaks. Never more than one per viewport
Silk drape	organic magenta radial-gradient blob behind imagery	hero, about
Arch crops	`border-radius: 200px 200px 16px 16px` on hero/category/gallery images, mixed with square crops	home, shop categories, PDP, about, 404
Oversized outlined numerals	Cormorant 600, transparent fill, 1.2px gold stroke (`-webkit-text-stroke`), 80–160px	look numbers, process steps, checkout sections, stats
Hang-tag prices	white tag with clipped point and punched hole (`clip-path` polygon + `::before` ring), gold border. `.big` , `.mag` variants	every price (cards, PDP, cart, delivery fees, admin)
Tape-measure size selector	gold tape with tick marks. Selected size = magenta segment. Sold out = struck-through	PDP, size guide, made-to-measure band
Tailor-label chips	square-ish label (radius 4) with an inner dashed border. Selected = ink with gold dashes	filters, topics, toggles
Sewn-button icon	magenta circle with 4 cream holes	cart count badge (with gold ring), toasts, confirmation seal, 404 "0", progress marker
Running-stitch loader	row of short magenta dashes animating left→right	loading states, uploads, "load more", MoMo approval wait
Needle / spool	small line icons. Spool illustration for the empty bag	empty states, made-to-measure

Editorial layout rules - Asymmetric hero: a large Cormorant headline (82px desktop / 50px mobile) with **one Great Vibes word** in magenta, overlapping photos (one arch, one stitched square), a hang-tag price on the photo, and an outlined look number. - Shop/home grids are **not uniform**: a 2×2 feature tile, staggered columns (±60px offset on desktop), quote tiles, and a mix of arch and square crops. - Craft storytelling blocks: **From sketch to stitch** (4 numbered steps), **Made to measure** (dark band with a tape measure and gold CTA), and an **Instagram/TikTok strip** (@sn_stitches_ / @sn_stitches). - Subtle Ghanaian textile references only as thin accents (kente band, wax-print photography). Never pattern-filled backgrounds.

Motion notes (all ≤ 400ms except the thread draw, and all respect reduced motion): - Product card hover: lift −6px, soft shadow, the photo gets a white stitched inset outline, a second image cross-fades, and "Quick add" slides up. - Thread draw on scroll. Hang tag swings 3° on hover. Add to bag: the sewn-button badge "pops" (scale 1→1.25→1) and a toast slides in. - Drawer slides in 350ms with a cubic-bezier(.2,.8,.2,1). The free-delivery stitch line grows. - Running-stitch loader: dashes shift 12px per 600ms loop.

Admin uses the same tokens but is calmer: an ink sidebar with a kente foot, white cards, stitched highlight only on the primary KPI and the customer card, tag prices, status chips, and a pipeline stepper. Logo: `assets/sn-stitches-logo.png` (transparent PNG, source) · `assets/logo.webp` (web, 240px) · `assets/favicon.png` .

Mood: **elegant modern African boutique**. Cream backgrounds, generous whitespace, editorial serif headlines, magenta used as the accent and CTA colour, gold for details (eyebrows, borders, dashed "stitch" lines). Never loud. The prints in the photography supply the colour.

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1. Design tokens

Colour

Token	Hex	Use
`magenta-600` (primary)	**#B0185E**	primary CTA, links, price on product page, active states, badges
`magenta-700`	#8C1149	CTA hover/pressed
`magenta-50`	#FDF3F7	selected radio/card background
`gold-500`	**#D4A437**	borders, dashed stitch dividers, focus ring, footer top border
`gold-700`	#B8891F	eyebrow text, wordmark "SN" (AA on cream for large text)
`ink-900`	**#141414**	text, secondary buttons, announcement bar, story section bg
`cream-50`	**#FBF6EE**	page background
`sand-100`	#F3EADB	footer, image placeholders, alt section bg
`line-200`	#E6DCCB	borders, dividers
`muted-600`	#6E655C	secondary text (AA on cream)
`white`	#FFFFFF	cards, inputs, drawers' footer
`success-600`	#1F7A4D	paid/delivered, success toasts
`whatsapp-600`	#1F8F4E	WhatsApp buttons only
`warning-500`	#C98A0B / bg #FFF8E6	pending, demo banner, low stock
`error-600`	#B3261E / bg #FDECEA	form errors, cancelled
`info-600`	#1F4E8C / bg #EAF1FB	processing, shipped

Status chips (admin): pending = warning, paid = magenta, processing = info, shipped = info (dark), delivered = success, cancelled = error, needs_attention = error outline.

Typography

- **Headings:** Cormorant Garamond 600 (500 for large display), line-height 1.1, slight negative tracking on display sizes.
- **Body / UI:** Jost 400/500 (300 for small captions), line-height 1.6.
- **Accent script:** Great Vibes, in magenta. Use it **at most once per view** (e.g. "stitched for you", "Thank you", "Stitch."), never for body text, prices or buttons.
- Eyebrows: Jost 500, 12px, uppercase, letter-spacing .22em, gold-700.
- Buttons: Jost 500, 13–14px, uppercase, letter-spacing .08em.

Style	Mobile	Desktop
Display (hero h1)	40px	60px
H1 page	34px	52px
H2 section	30px	44px
H3 card/product name	19px	20px
Body	16px	16px

Style	Mobile	Desktop
Small	13–14px	14px
Price (PDP)	22px magenta	24px

Spacing, radii, elevation

- 4px base scale: 4, 8, 12, 16, 20, 24, 32, 40, 56, 72, 96. Section padding 56px mobile / 72–96px desktop. Container max 1200px, gutters 20px (mobile) / 32px.
- Radii: inputs 10px, cards/images 14px, chips and buttons **pill (999px)**, drawers 0.
- Shadows: minimal. Drawer `0 0 40px rgba(0,0,0,.12)`. Cards are flat, and hover zooms the image instead of adding a shadow.
- Motion: 200–350ms ease-out. Respect `prefers-reduced-motion`.
- Signature detail: **dashed magenta or gold "stitch" lines** (2px dashed) as dividers, echoing the logo badge.

Breakpoints (mobile-first)

`sm 480` · `md 768` · `lg 1024` · `xl 1280`. Design at 360–390px first. Product grid 2 cols < 768, 3 cols ≥ 768, 4 cols ≥ 1024.

2. Components

- **Announcement bar:** ink bg, cream text 12px, one line, scrolls on mobile if long. Editable in admin.
- **Header (sticky, 64px mobile / 72px desktop):** cream 94% with blur and a bottom border. Mobile: hamburger · centred logo (48px tall) · bag icon with magenta count badge. Desktop: logo + "SN *Stitches*" wordmark left, nav right (Home, Shop all, Ankara, Kente, Evening, Casual, Our story, Contact), search icon [P2], bag. Nav links get a magenta underline that animates on hover.
- **Mobile menu:** full-width sheet below the header, 48px tap rows, socials row at the bottom (IG, TikTok, WhatsApp).
- **Buttons:** Primary (magenta fill, white text). Secondary (ink fill). Ghost (1.5px ink outline). WhatsApp (green with WA icon). Text link (magenta, underline). Sizes: 48px tall (default, a good tap target), 40px small. Disabled: 50% opacity, no hover. Loading: spinner + "Processing…".
- **Product card:** 3:4 image, radius 14. The second image crossfades on hover (desktop). "View dress" pill slides up on hover. Badges top-left: Sale (magenta), New (ink), Sold out (muted), Sample (gold outline, admin/preview only). Below: name (serif 19–20), meta "ANKARA · MIDI" (caps, muted), price (strike-through compare-at).
- **Category tile:** 4:5 image, bottom gradient, serif name and gold small-caps count.
- **Chips (filters):** pill, white with a line border. Selected = ink fill. 36px tall.
- **Size selector:** 52×46 rounded-10 buttons XS–XXL. Selected = ink. Out of stock = 35% opacity with strike-through and `aria-disabled`, tooltip "Sold out — WhatsApp us to pre-order".
- **Colour swatch:** pill with a 24px circle and the colour name.
- **Quantity stepper:** pill, − / value / +, 1–10.
- **Inputs:** 48px, white, line border, radius 10. Label above (13px muted). Focus = 2px gold outline. Error = error border + message below with an icon. Phone input shows a 🇬🇭 +233 prefix hint.
- **Radio cards:** delivery and payment choices, full width, magenta border + magenta-50 bg when selected, price right-aligned.

- **Payment logos row:** MTN MoMo (yellow), Telecel Cash (red), AirtelTigo Money (blue), Visa/Mastercard. Small and honest, no fake "secured" claims beyond "Payments by Paystack".
- **Cart drawer:** right sheet, full width on mobile, 420px on desktop. Line items (76px thumb, name, size · colour, stepper, remove, line total), free-delivery progress bar (gold), subtotal, Checkout (primary) and Order via WhatsApp (green). Escape/scrim closes it. Focus is trapped.
- **Toast:** ink pill at bottom centre, 2.5s, `role=status`.
- **Accordion:** serif summary with +/− (Details, Fabric & care, Delivery & returns).
- **Skeletons:** sand-100 blocks with a shimmer for cards, PDP and table rows.
- **Empty state:** script line ("Your bag is empty") + short text + ghost CTA.
- **Footer:** sand bg, gold 3px top border. Logo + tagline "FASHION · DESIGN · STITCH", columns Shop / Help / Studio. Socials: Instagram **@sn_stitches_**, TikTok **@sn_stitches**, WhatsApp/Call **054 692 3596**. Legal links (Privacy, Terms, Returns).
- **Floating WhatsApp button** (mobile, bottom-right, 56px, green). Hidden on checkout and while the drawer is open.

3. Storefront pages

Home (`/`) — mockup 01 (the mockup wins over the text where they differ)

1. Announcement bar → header.
2. **Hero:** campaign image right/background with a cream gradient on the left on desktop; image on top and text below on mobile. Eyebrow "NEW SEASON · MADE IN ACCRA", H1 "Dresses with a story, *stitched for you*" (script line), one-sentence copy, CTAs "Shop the collection" (primary) + "Evening edit" (ghost).
3. **Featured dresses:** 4 cards (2×2 mobile) + "View all →".
4. **Shop by category:** 4 tiles.
5. **Our story teaser:** ink section, image + text, "Fashion. Design. *Stitch.*", dashed magenta stitch divider, CTA "Our story".
6. **Value props:** Delivery across Ghana · Pay with MoMo or card · Perfect fit (XS–XXL + made-to-measure). Cards with a gold dashed border.
7. **Instagram/TikTok strip:** "Follow @sn_stitches_" with 6 images linking to IG (P1 static images, P2 feed).
8. Footer.

Shop (`/shop`, `/shop/[category]`) — mockup 02

- Page head: eyebrow + H1 + short category description.
- Desktop: left filter rail 240px. Mobile: "Filters (n)" button opens a bottom sheet with "Show N dresses" and "Clear".
- Filters: Category (chips), Size (chips XS–XXL, matches products with that size in stock), Price (range slider + presets "Under GH₵400 / 400–800 / 800+"), Length (Mini/Midi/Maxi), In stock only (toggle).
- Toolbar: result count, sort select (Featured, Newest, Price low→high, high→low, Name A–Z).
- Filters reflect in the URL query (shareable, back button works). Grid with skeletons while loading. Empty: "No dresses match those filters" + Clear filters. Pagination: "Load more" (24 per page).

Product (`/product/[slug]`) — mockup 03

- Desktop 2 columns (gallery 55%). Gallery: vertical thumbs + main 3:4 image, click to zoom/lightbox. Mobile: swipeable carousel with dots, thumbs below.

- Right: breadcrumbs, eyebrow (category · length), H1, price (magenta) + compare-at, short description, Colour swatches, Size selector + "Size guide" link (opens a modal with the table), stock hint ("Only 2 left in M"), quantity, **Add to bag** (primary, full width on mobile, **sticky bottom bar on mobile** once scrolled past it), "Order this on WhatsApp" text link, shipping line "Ships in 1–3 days · Accra same-day available".
- Made-to-order: label "Made for you in N days", plus a measurements note.
- Accordions: Details, Fabric & care, Delivery & returns.
- "You may also love": 4 cards from the same category.
- Validation: Add without a size gives an inline error "Please choose a size" and focuses the size group.

Cart drawer — mockup 04

As in the component spec. Prices re-validated on open (server action). If an item is no longer available: inline warning on the line and the Checkout button disabled until it's removed. The free-delivery bar reads "Add GH₵600 more for free delivery" / "You've unlocked free delivery 🎉".

Checkout (`/checkout`) — mockups 05 & 06 (delivery step, then payment step with MoMo network cards)

- Single page, guest checkout. Mobile order: collapsible **order summary** at the top ("Show order summary · GH₵940"), then sections. Desktop: form left, sticky summary right.
- Sections: **1 Contact** (full name, phone required + "We'll call/WhatsApp this number", email required for the receipt). **2 Delivery** (region select with all 16, city/town, GhanaPost GPS optional, address & landmark textarea, delivery option radio cards that appear after the region is chosen, with price/ETA; same-day greys out after 12:00 with the reason). **3 Payment** (radio cards: Mobile Money [MTN, Telecel, AirtelTigo logos, "Approve on your phone"], Card, Order via WhatsApp [pay on confirmation]). Optional discount code [P2]. Order note (optional).
- CTA reflects the method: "Pay GH₵940 with Mobile Money" / "Pay GH₵940 by card" / "Send order on WhatsApp".
- Errors: inline per field + a summary at the top. Focus goes to the first error. Ghana phone validation (`0XXXXXXXXX` or `+233XXXXXXXXX`).
- States: submitting (button spinner, form disabled), Paystack popup open, popup closed ("Payment not completed — try again or choose another method"), payment failed, config missing (warning banner, in dev only).

Confirmation (`/order/[orderNumber]`) — mockup 07

Magenta check circle, eyebrow "ORDER SNS-24001", H1 "Medaase, Akosua! *Thank you*". Copy: we'll call/WhatsApp on the phone number, receipt sent to the email. Summary box (items, delivery, total, status, Paystack ref, ship-to). If the status is still pending: "Confirming your payment…" with a poll every 3s up to 60s, then the message "We're confirming with your network — you'll get an email shortly". CTAs: Continue shopping · Chat with us on WhatsApp (prefilled with the order number).

About / Our story (`/about`) — mockup 08

Editorial: hero image, the story of SN Stitches (Accra studio, hand-finished, wax print and kente), "Fashion · Design · Stitch" values in three columns, studio photos, CTA to shop. **Copy is placeholder for the owner to supply.**

Contact (`/contact`) — mockup 09

Cards: **WhatsApp** (button → `wa.me/233546923596`), **Call** `tel:+233546923596` (054 692 3596), **Instagram** @sn_stitches_ (instagram.com/sn_stitches_), **TikTok** @sn_stitches (tiktok.com/@sn_stitches), email [placeholder]. Studio hours (Mon–Sat 9am–6pm, assumption) and location (Accra; exact address TBC). Simple contact form [P2] (stores a message and emails the owner).

Size guide (`/size-guide`, also as a modal) — mockup 10

Table XS–XXL × UK size, bust, waist, hips (inches, cm toggle), how-to-measure illustration, "Between sizes? Size up, or WhatsApp us for made-to-measure."

FAQ / Delivery & returns (`/faq`) — mockup 10

Accordions grouped: Ordering & payment (MoMo, card, WhatsApp, is it safe), Delivery (zones, fees, cutoffs, free threshold, pickup), Sizing & made-to-measure, Returns & exchanges (7 days, unworn, tags on, exchange or store credit; **policy is an assumption to confirm**), Care. FAQPage JSON-LD.

4. Admin pages (`/admin`)

Admin UI uses the same tokens but is denser: white surfaces, 14px body, ink sidebar on desktop, bottom tab bar on mobile (Dashboard, Orders, Products, Settings). It must work well on the owner's phone. - **Login:** centred card on cream, logo, email + password, "Send magic link" alt, errors inline. Non-admin users see "You don't have access." - **Dashboard:** KPI cards (Today's sales, This week, Orders to process, Low stock), "Needs attention" list (paid but not processed > 24h, stock conflicts), recent orders table, quick actions (Add product, View pending). - **Products list:** search, category/status filters, table on desktop (thumb, name, category, price, total stock, status, featured) and cards on mobile. Bulk archive [P2]. "Sample" badge. - **Product editor:** two columns on desktop. Left: name, slug (auto), description (textarea/markdown), details bullets, category, length, price, compare-at, made-to-order + lead days. Images: dropzone, thumbnails reorderable (drag), alt text, set main, delete. **Variants grid:** colours × sizes, stock number inputs, row/column quick-fill, active toggles. Right: status (Draft/Active/Archived), featured toggle, SEO title/description with a Google preview. Sticky save bar with unsaved-changes guard. Toast on save. - **Orders list:** status tabs with counts, search, date range, table (order #, date, customer, phone, region, total, payment method, status chip). Tap a row to open the detail. - **Order detail:** header (order # + status chip + date). **Pipeline stepper** Pending → Paid → Processing → Shipped/Out for delivery → Delivered with the next-action button ("Mark processing", "Mark shipped" with an optional courier/tracking note) and "Cancel order" (confirm dialog, restock). Items with images, sizes and qty. Customer card with **WhatsApp** and **Call** buttons and a copy-address button. Payment card (method, channel, Paystack ref, verified time). Timeline of events and internal notes. Print packing slip. - **Settings:** store contacts/socials, announcement text, free-delivery threshold, delivery options table (label, fee, ETA, cutoff, active).

5. States

State	Pattern
Loading	skeletons (no spinners for page content). Button spinners for actions
Empty	script headline + one line + CTA (bag, search results, orders list "No orders yet")

State	Pattern
Error	inline field errors. Page-level error card with "Try again" + WhatsApp help link. Custom 404 "This dress has wandered off" with links to shop
Offline / slow	images have sand placeholders and blur-up. Cart works offline. Checkout shows "You're offline" if `navigator.onLine` is false
Sold out	card badge, PDP sizes disabled, "WhatsApp to pre-order"

6. Accessibility

- WCAG 2.2 AA: text contrast ≥ 4.5:1 (magenta #B0185E on cream ≈ 6.4:1 ✓, gold only for large text and decoration; use gold-700 for small eyebrow text).
- All interactive targets ≥ 44×44px. Visible gold focus ring. Skip-to-content link.
- Semantic landmarks, one H1 per page, form labels tied to inputs, `aria-live` for cart count and toasts and errors, `aria-pressed` on size/filter chips, radio groups use fieldset/legend.
- Drawer/modal: focus trap, Escape closes, focus returns to the trigger, `inert` background.
- Alt text describes the dress (colour, print, cut). Decorative images use `alt=""`.
- Prefers-reduced-motion disables zoom/crossfade. Language `en-GH`. Prices read as "GH₵450" with `aria-label="450 Ghana cedis"`.

7. Content & formatting

- Currency: `GH₵1,250` (no decimals unless needed). Ghana phone display `054 692 3596`. Dates `29 Oct 2026`, times in Africa/Accra.
- Tone: warm, confident, a little Ghanaian ("Medaase", "Akwaaba"), short sentences.
- All seed products, prices and photos are **samples**. Mark them in the admin and replace them before launch.

Admin · Add new listing (`/admin/products/new`), mockups 27 & 28

This page is separate from the product editor (24). Open it from the **"+ Add new listing"** button on the Products list (23). - **Stepper:** 01 Photos → 02 Details → 03 Pricing → 04 Sizes & stock → 05 Preview → 06 Publish. Desktop shows the steps as numbered stitched cards on one page. Mobile shows a compact stepper and one step at a time. - **01 Photos:** drag/drop, file picker or camera (`capture="environment"`) upload to Vercel Blob. Drag to reorder, star the main photo, alt text required for each photo. "Generate studio shot" is shown disabled with a *Phase 2* tag. - **02 Details:** name; slug auto-generated (editable, uniqueness checked); description; category; length/style (DressLength incl. PLAYSUIT); fabric; care; tags. - **03 Pricing:** price in GH₵ (stored as pesewas), optional compare-at (must be greater than price), made-to-measure toggle plus GH₵ add-on fee. - **04 Sizes & stock:** grid of colours × XS–XXL with quick-fill ("all = 2", "clear"). - **05 Live preview:** product card (hang-tag price) plus a phone mock of the product page; updates live. - **06 Publish checklist:** ≥1 photo with alt text, name, price, category, ≥1 size in stock. **Publish** stays disabled until every item passes. **Save draft** is always available. - **Autosave:** draft saved every 10 s and on blur ("Saved · 12:04"). - **Mobile:** sticky bottom bar with "Save draft / Next: →"; topbar actions hidden. - **28 Success:** seal plus "Listing published!", hero photo with price tag. Actions: View in shop, Share to Instagram @sn_stitches_ (download image + copy caption), Share on WhatsApp (`wa.me/?text=`), TikTok, copy link. Also: Add another listing / Edit / Back to products.